

Chapter 4

Sound Production in Finished Instruments



Chapter Introduction

As it has been described in 3 earlier chapters this chapter is about improving the dynamic and acoustic outcome of the described structure.

What follows is an exploration of the structural principles that allow the violin to activate itself when the played frequency aligns with its internal stress conditions. This perspective not only clarifies long-standing misunderstandings, but also brings us closer to the original intention behind the instrument's design: a self-activating system capable of producing a coherent, balanced, and remarkably precise spectrum of frequencies.

What the Luthier Can Do to Improve Sound Production on STL-Shaped Instruments.

In the previous chapters, the motor system and the geometric form of the violin were described in detail. This chapter builds on that foundation by showing how the luthier can refine these structural conditions to improve the instrument's dynamic and acoustic outcome.

What follows is an exploration of the structural principles that allow the violin to activate itself when the played frequency aligns with its internal stress conditions. This perspective not only clarifies long-standing misunderstandings, but also brings us closer to the original intention behind the instrument's design: a self-activating system capable of producing a coherent, balanced, and remarkably precise spectrum of frequencies.

No matter how skilfully an instrument is built, we cannot assume that its structural state is already optimal. It must be calibrated — and such calibration can only be performed on a playable instrument.

We now turn to the structural behaviour that produces the dynamics of the Motor System, as shown in Figure 1.5. On the top plate, two structural regions buckle outward while a central region bends downward. On the back plate, the structure bends upward over the soundpost. This behaviour becomes possible because the end blocks are pulled upward and inward by string tension. All of these movements occur relative to a non-moving column: the soundpost. Without this column, the system cannot function.

The structures of the top and back plates interact as a tandem system, as described in Chapter 1. The task here is to bring that tandem system into its optimal dynamic behaviour. When this occurs, sound emerges, stabilises, and becomes adjustable by the luthier.

Because the structure and its primary function have already been explained in earlier chapters, it is important that the next steps are presented in a way that is both easy to understand and practical to apply. In this chapter, we go further: we examine what can actually be done with the function of the structure, since we cannot expect the wood by itself to respond correctly to the action of the strings. Even when explained simply, the structure remains a complex system.

This chapter therefore describes the aspects the luthier can influence directly, beginning with the most important task: improving the Motor System. The focus now shifts from construction to function — how the assembled violin, complete with soundpost, bridge, and strings, produces its characteristic acoustic output.

To investigate this, two instruments were tested for their sound behaviour by playing the open G string and recording the response with Audacity. Despite differences in the wood,

the resulting frequency spectra are nearly identical. The structures with internal framework are equal. Both instruments exhibit the BH peak (around 2000 Hz) as well as the characteristic frequencies associated with constructive interference.

We begin by examining the response obtained when strings are placed on the instrument for the very first time. Figures 4.1 and 4.2 show the recordings from the two instruments, presented one above the other. What we see is the acoustic result of the open G string being played.

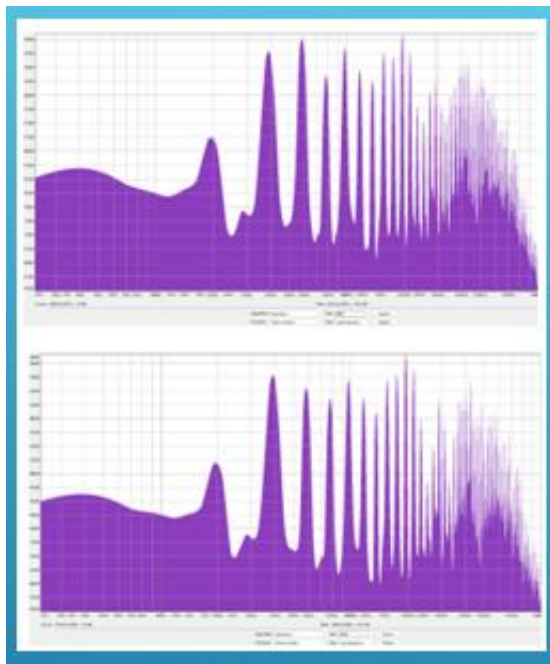


Figure 4.1 2048 FFT resolution

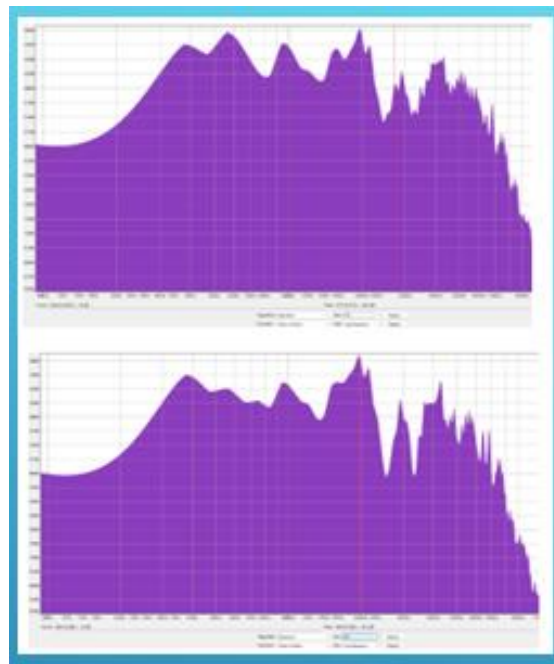


Figure 4.2 512 FFT resolution

When the STL framework conditions are fulfilled, the violin behaves as a mechanically governed system. This was not an accidental by-product of craftsmanship, but the hidden functional intention of the original inventor. The geometry was conceived so that, once assembled, the instrument would **operate as a mechanically optimised sound-producing device.**

The genius of the design lies in the fact that the wood itself does not need to “understand” how to respond. The geometry and the internal framework impose the correct behaviour. The plates, blocks, and soundpost are arranged so that the forces generated by the strings are channelled into a predictable sequence of structural reactions. When these reactions occur in the correct order and proportion, the violin becomes a self-regulating mechanical system: stable, efficient, and capable of producing a consistent acoustic output.

For centuries, this principle remained hidden because the system reveals itself only when the STL conditions are met. Without the correct internal framework, the instrument

behaves like ordinary wood under load — unpredictable, inconsistent, and dependent on material variation. With the STL framework in place, however, the instrument transcends the variability of the wood. It becomes a governed mechanism, one whose dynamic behaviour is defined by geometry rather than by material chance.

This insight changes the role of the luthier. The task is no longer to hope for favourable wood or to rely on intuition alone, but to bring the structure into its intended mechanical state. Once this is achieved, the violin behaves exactly as its inventor designed it to behave.

The luthier's task is therefore not only to build the structure that activates its mechanical function. Once the violin is assembled — soundpost, bridge, and strings in place — the luthier can improve the instrument's sound not only by adjusting these free, movable components. Their position and tension determine how effectively the underlying STL geometry can express its mechanical behaviour. At this stage, the luthier must trim and refine the mechanical-dynamical function of the violin structure.

However, when an instrument has been built according to the STL principles described in Chapters 1–3, the luthier's role becomes both more powerful and more precise. The structure already contains the potential for optimal function. What remains is to bring the system to life by activating its internal mechanisms:

- the motor function,
- the dipole function,
- and, when possible, the interference function.

When the STL framework conditions are fulfilled, the violin behaves as a mechanically governed system. This was not an added feature but the hidden functional intention of the original inventor. The geometry was designed so that, once assembled, the instrument would operate as a mechanically optimised sound-producing device. These functions are not added to the instrument — they are unlocked through correct adjustment.

The luthier's work at this stage is therefore not guesswork or intuition alone, but the deliberate optimisation of a mechanical system designed to respond predictably when placed in its optimal condition. The inventor's solution is nothing short of ingenious: a structural logic that remains dormant until the instrument is brought into the precise state for which it was conceived. Once this state is achieved, the violin reveals its true nature.

The plates, blocks, and soundpost no longer behave as independent pieces of wood but as components of a unified mechanism. The system becomes self-regulating, stable, and acoustically efficient. What appears to be craftsmanship becomes, in fact, the activation of a mechanical principle embedded in the geometry itself.

This insight reshapes our understanding of the violin. It shows that the instrument is not merely a product of fine woodworking, nor a passive resonator waiting to be excited by the strings. It is a designed mechanism — a dynamic structure whose behaviour can be predicted, calibrated, and optimised. The luthier's task is to reveal this mechanism, to bring the structure into the condition in which its intended function becomes active.

4.1 The Central Question: What Can Be Done to Improve the Function of the Instrument

Having explored the structural revelations presented in Chapters 1–3, we now arrive at the essential question that guides the entire investigation:

- **Where and how does sound originate within the violin?**
- **How can the outcome of sound production be improved?**

These questions are not merely academic — they define the practical task of the luthier working with a finished instrument. Understanding the origin of sound and the mechanisms that shape it is the foundation for any meaningful acoustic adjustment.

This chapter provides answers based on the specific structural qualities revealed earlier, and it explains what further actions can be taken to control and improve the acoustic results.

4.2 What Is Important Besides the Geometric Structure

Beyond the geometric structure described in Chapters 1, 2, and 3, the next essential element is the function of the Motor System, introduced earlier in the manuscript. Geometry alone provides the conditions for mechanical behaviour, but it is the Motor System that transforms these conditions into sound.

For this reason, we begin by observing the sound that emerges when the bow activates the strings — especially the open G string. This string provides the clearest window into the instrument's fundamental behaviour and into the operation of its Motor System. Its low frequency, long wavelength, and strong interaction with the body make it ideal for revealing how the Motor System initiates and sustains vibration.

The Motor System initiates the main dynamics of the tandem system and maintains these dynamics, as described in Chapter 1. By studying the response of the G string, we can identify how the instrument's internal mechanics begin to operate, how energy is

transferred into the body, and how the structural conditions established during construction influence the resulting acoustic output.

Once activated, the structures begin vibrating in relation to the STL framework, which directs where and which structural regions become active. This does not mean that an STL line cannot vibrate; rather, it distributes the main vibration so that the energy remains concentrated in the bulging arching shapes, where the dynamic behaviour is most effective.

To obtain initial information, a simple recording program can be used. The resulting graph reveals the produced frequencies. Figures 4.1 and 4.2 show the spectrum of the played open G string. It is immediately clear that many frequencies other than the fundamental 196 Hz appear — frequencies that are difficult to hear individually but are nevertheless present. These results indicate the quality of the produced sound and the relative dB levels at which the various frequencies occur.

4.3 Why the G String?

All four strings carry approximately equal tension. This is not only to balance the forces acting on the instrument, but also for another essential reason: the G string is the most flexible of the four strings, and when played it influences the Motor System more strongly than the others. Because of its flexibility, the G string releases and activates the structural force action more than any other string.

The changing stress from the G string acts directly on the end blocks and initiates the three-beam system to become dynamic. Once this system begins to function, it in turn activates the spring functions of the bout structures. In other words, the G string is the most effective trigger for revealing the true quality of the instrument.

If the Motor System does not operate correctly, the result is a lifeless instrument — one that cannot respond dynamically, cannot develop dipole function, and cannot express the structural potential built into its geometry. To make it easier to follow the next description of what happens here, Figure 3.7 is shown again.

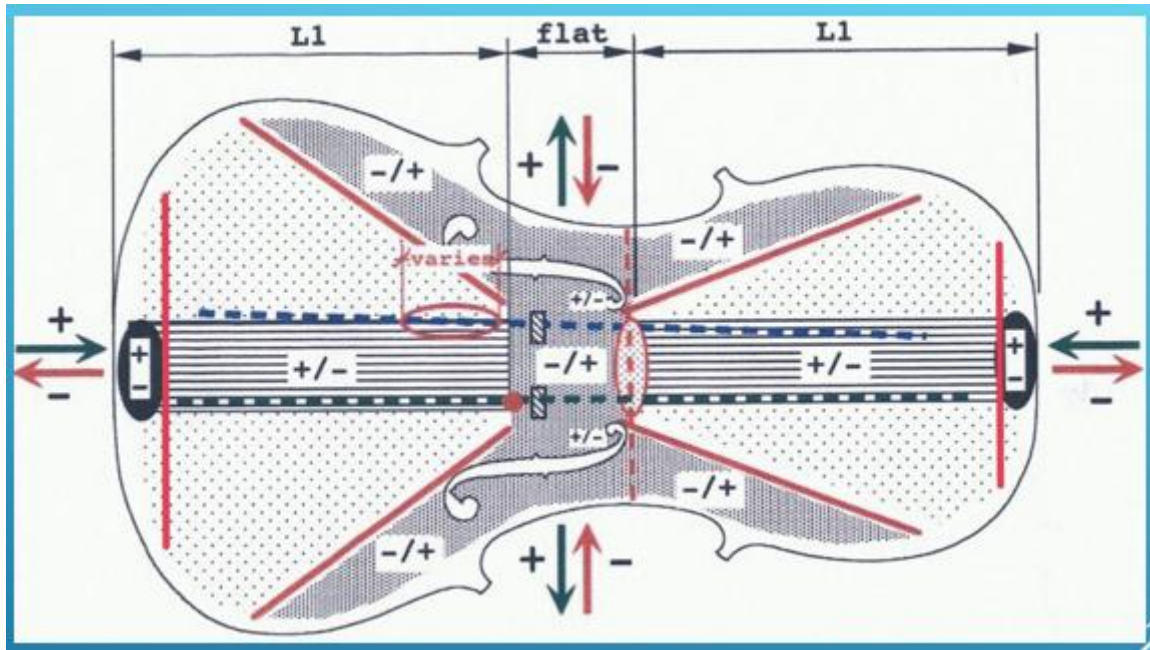


Figure 3.7

The three beams extend from end block to end block, with the “island” located in the centre. Together with the back plate, this structure forms the **motor** of the instrument. The two beams on each side of the island are equal in length and have similarly shaped arching curves, although their thickness graduation may differ depending on how the luthier has shaped them. This report deals with these issues and explains how the motor function can be optimized.

The adjacent structures around the beams are indicated by the dotted area. The STL angles relative to the centreline vary, creating differently shaped bout structures. These adjacent wood regions contain a large number of parallel winter-growth beams.

These beams must become controlled by their function, which arises from their buckling behaviour; as this occurs, they develop spring-like qualities. Their lengths differ: they begin at the vertical red line shown in Figure 3.7 and end at the red STL line. Each beam has its own curvature and its own chord line.

Because of this, they possess highly specific structural qualities in relation to how buckling stress acts on them, and therefore they produce different responses. Their structural quality must generate spontaneous stress reactions that correspond to

specific played frequencies. Thus, there is a direct relationship between frequency and the stress-response quality of these beams.

4.4 Luthiers trimming of free plates

Many luthiers use the technique developed by Carleen Hutchins, vibrating the plates to identify the M2 and M5 modes. These modes inform the luthier about the stiffness of the structure and help guide thickness graduation. It is a sensitive process, and it strongly influences the formation of the node lines. Figures for M2 and M5 shown here are taken from a violin under construction.



M2



M5

What is important is that the layout of the M2 and M5 modes also appears when the instrument is under **string load**, as shown in Figure 4.1. In that figure, we see two Mode-5 node lines and, essentially, the Mode-2 pattern.

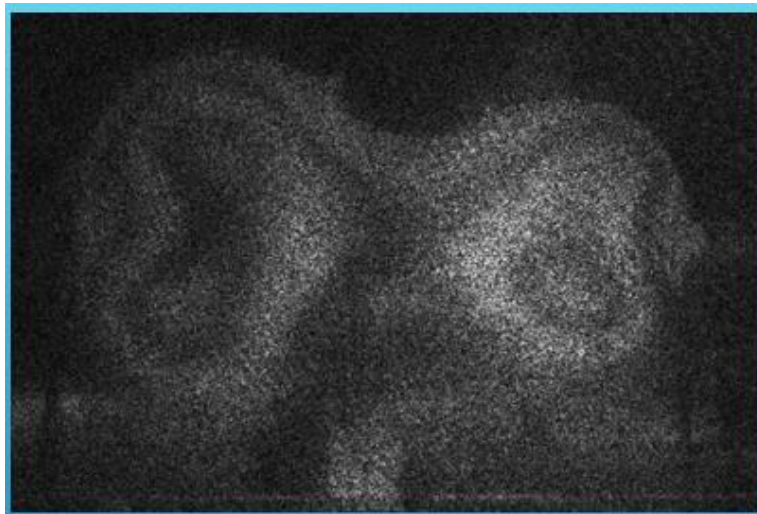


Figure 4.1

The two Mode-5 node lines form a heart-shaped figure, which coincides with the structure of the STL framework shown earlier.

When Carleen Hutchins shaped her plates, she did not know that this rare structural quality would appear in the assembled instrument under string load. Nor was she aware of the structural significance of special arching shape conditions.

The experienced maker simply needs to look and "eye-ball" the shape of his arches, for he knows how to change the arches relative to the grain of the wood and various other subtleties that cannot be got from any sort of measurement technique. The arch of the top has a shape that is

She wrote that the experienced maker must rely on long practice and visual judgment, because **no measurement technique** can fully describe the subtleties of the arch.

What Hutchins expresses is essentially this: ***we do not know anything for certain, but with long experience we may succeed.***

Hutchins — like all luthiers and scientists after her — had no information about the geometric structure that governs the functional behaviour of the instrument.

Figure 4.2.2 — Structural Logic Revealed in the Back Plate

The back plate, shown in figure 4.2.2, shows the STL lines and thickness values that define the instrument's internal beam structure.

These STLs form the basis of an **octahedral framework**, and their precise alignment with the outer arching demonstrates intentional geometric design.

This is the physical foundation of the violin's **frame work system**: a coordinated structure prepared to activate through spring behaviour on arching shape when placed under string load.



Figure 4.2.2

Figures 4.2.3 and 4.2.4 show how the STL in and out side measured on one going work in its final stage a line above each other some with minor differences.

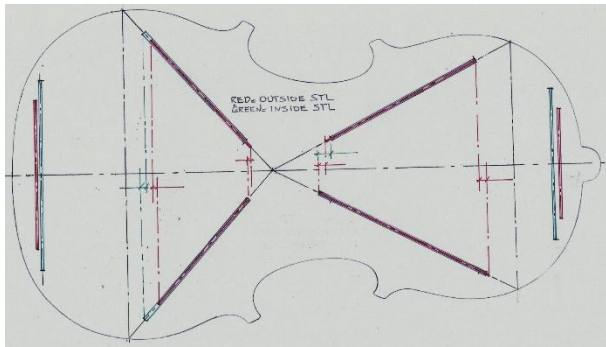


Figure 4.2.3 back plate

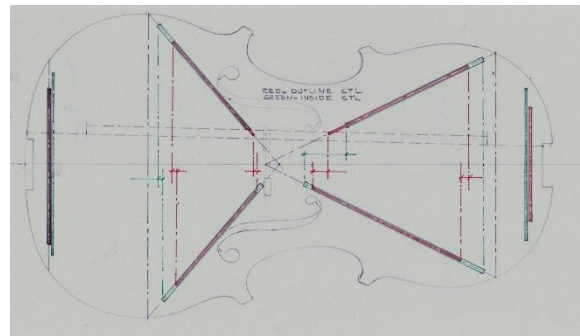


Figure 4.2.4 top plate

Yet this information is crucial, because it strongly assists the luthier in making a successful instrument.

The following figures show that the exact arching geometry described in Chapter 2 can produce **multiple Mode-5 conditions**, of which **M5.2** represents the most commonly observed frequency.



M5.1



M5.2



M5.3

The results from the holographic investigation show that the **internal framework based on the STLs will** influence the vibration patterns, as visualised in Figure 4.1.

What is now understood to be important is the **nodal shape of Mode 5**, the curve commonly referred to as the ring mode. The structures on both sides of this nodal line move either upward or downward.

Therefore, it is essential to understand about **inflection point**—the location where the movement reverses direction—governs the behaviour of the buckling bout shape, and thus the function of the motor system.

The **inflection points building a line** is the structural boundary where the plate reverses its direction of movement. On one side of the line, the plate bulges outward; on the other side, it bends downward. This reversal is the key to understanding how the violin's motor system operates when active.

A similar important result is found on the top plate, although no photographs were taken because the fingerboard and tailpiece cast shadows over the plate structures.

4.5 The behaviour on the beam structure.

Figure 4.5 shows how the string load acts on the end block and produces buckling behaviour. We see the critical inflection point where outward buckling and downward bending arise in front of the end block.

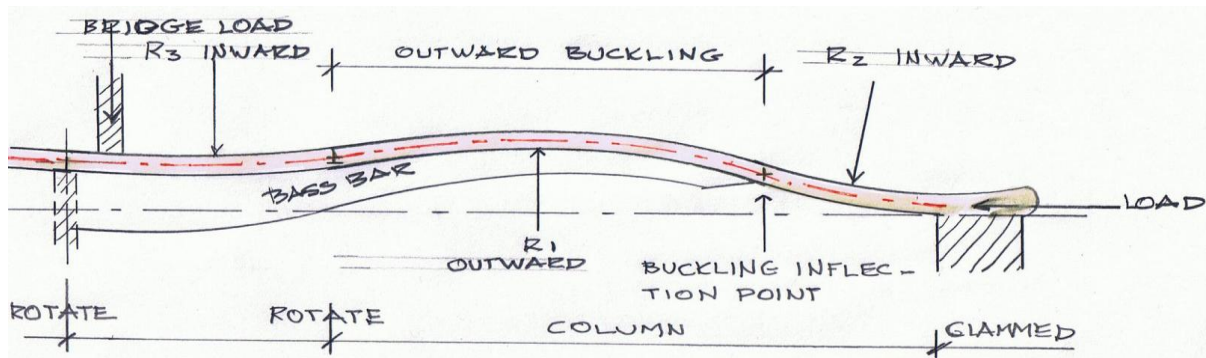


Figure 4.5

Figure 4.1 shows that an inflection line forms along the outline of the instrument and then transitions into a straight line at the location of the STL framework. What we observe in Figure 4.5 is the transition between outward buckling at the inflection point and downward bending near the end block, as well as how the bridge load forces the beam on the “island” downward.

The downward bending of the bridge on the “island” deforms this region in two directions. One deformation corresponds to what is shown in Figure 4.5; the other is a transverse flattening.

As the island flattens, the chord line beneath it becomes longer. The resulting stress condition remains linked to the stress conditions in the surrounding bout shapes.

Thus, a balance exists between the stresses in the bout shapes and those acting on the island..

This interaction produces the dynamic process generated by the motor system — the mechanism that gives rise to both the dipole function and the breathing quality.

This behaviour can be controlled at specific locations along the beam structure.

At these locations, the stress conditions acting on the bout arching become governed and controlled. Without such defined structural points, it is impossible to understand the internal behaviour of the bout and C-bout. The behaviour can differ at four locations on the top plate as well as on the back plate, and without structural guidance a dynamic system arises that may not remain in balance.

Section 4.4 describes how a well-shaped system influences the function of the motor system and what the luthier can do to improve its performance. Earlier, the geometry was shown to form an octahedral configuration (Figure 4.6), a structure that enables the luthier to influence the dynamic behaviour of the arching shapes.

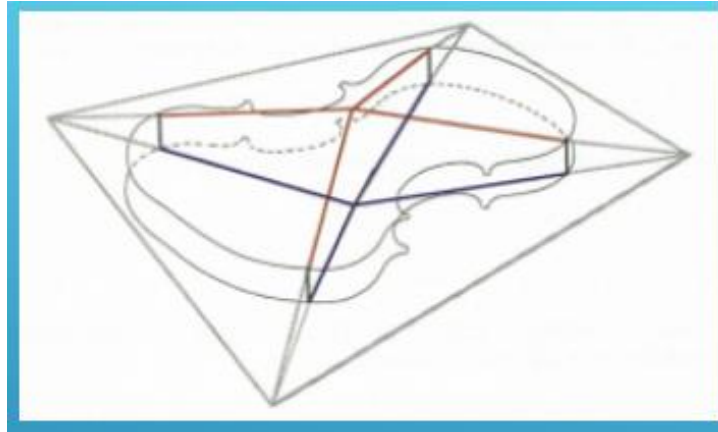


Figure 4.6

The STL framework defines the internal geometry that governs how the plate can deform. Each STL line acts as a mechanical guide rail:

- it limits rotation
- it redirects stress, and
- it divides the plate into regions that respond differently to string load.

When the instrument is under tension, the **inflection line** naturally forms along the outline of the instrument, because this is the region where the plate is least constrained. As the deformation approaches the STL lines, the behaviour changes: the inflection line straightens and aligns with the STL geometry. This occurs because the STL lines are stiffer in its octahedron framework and resist curvature, forcing the plate to rotate **around** them rather than **through** them.

This interaction explains why:

- the bout areas rise upward
- the C-bout is pressed downward and
- the motor system can only function if the inflection line is free to form at the correct location.

If the STL geometry is incorrect, or if the bass bar crosses the functional inflection point at its end, the system becomes blocked. The plate cannot reverse its movement cleanly, the bout cannot buckle, and the motor system loses its ability to drive part of the instrument.

Figure 4.5 shows the longitudinal cross-section along the centreline of the upper bout.

A similar condition is observed in the lower bout, particularly in relation to the end block and the sound post.

As seen in Figure 3.7, there is no fixed pivot near the bridge feet like the sound post.

This is fundamentally a structural problem that requires a solution.

The bass bar provides part of this solution: it prevents outward buckling along the f-hole region and introduces a stabilising structure, although not a fixed pivot.

The STL and the bass bar together determine the stress distribution and dynamic behaviour.

This area is well known to luthiers, especially in relation to controlling what is commonly called the wolf-tone problem.

The region is often too flexible, and various devices are sometimes added to damp the dynamic behaviour near the f-hole stabilizing stress condition.

Figure 4.5 shows an exaggerated sketch of the deformation to make the structural behaviour visually understandable. In reality, the deformation is extremely small—less than 0.05 mm.

We observe the following sequence of structural pivots and transitions:

- the **sound-post pivot transition point**,
- the **bridge feet**, producing downward load,
- the **upper-hole pivot transition point**,
- the **zone of outward buckling (R1)**,
- the **zone of inward bending (R2)**,
- the **island, showing downward bending (R3)**,
- and the **end block**, which moves inward and upward.

Behind this region lies the bass bar, whose end must not cross the functional inflection point. If it does, it restricts the essential up-and-down movement of the inflection point and thereby blocks part of the dynamic function of the motor system.

4.5 Acoustic Implications of Buckling Behaviour

The **buckling of the central beam system** is not merely a structural deformation; it is the **mechanical origin of dynamic breathing and interference control** in the violin.

- When the **upper and lower bout beams buckle outward** and the **island structure buckles inward**, the instrument undergoes alternating phases of **compression and expansion**.
- These phases **modulate internal air pressure**, creating the breathing motion that drives **acoustic radiation**.
- The **parallel response of top and back plates** ensures that energy is distributed symmetrically, stabilizing the vibrational field.
- The **F-hole slits** allow the centre of the top plate to move downward, enabling the formation of the **three-beam system**. This geometry is critical for producing **interference conditions**, where multiple wave sources converge and enrich the frequency spectrum.

In essence, the **buckling logic transforms static stress into dynamic acoustic output**. The violin's body does not simply resist string tension; it converts that tension into a **controlled breathing mechanism** that governs how interference patterns arise and how sound is projected outward. The spring function of the bout shapes is an integral part of the motor system.

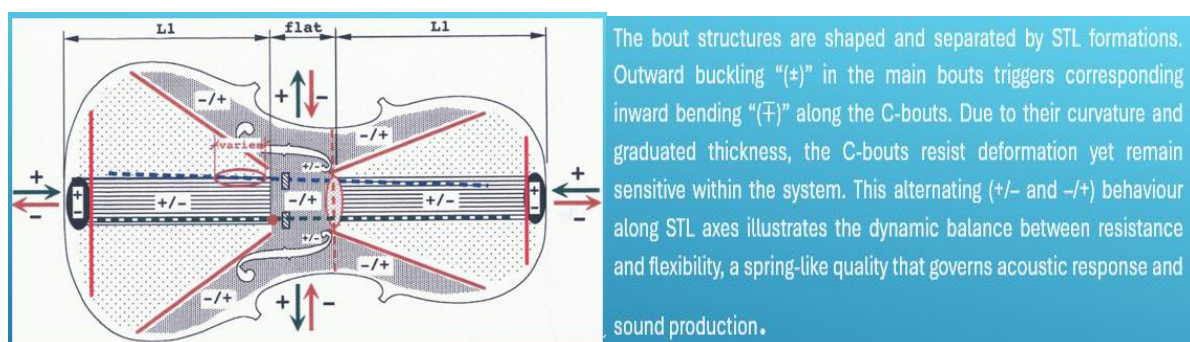


Figure 4.7

4.6 Extended Analysis: STL Bout Behaviour and Acoustic Implications

Building on the behaviour illustrated above, we can examine how stiffness reduction, creep, and spring-like qualities influence the violin's acoustic response:

- The C-bout areas, shaped by curvature and graduated thickness, are naturally stiffer. This stiffness resists deformation but remains highly sensitive within the system.

- When stiffness is reduced, the deformation behaviour of the main bout shapes changes, altering both their dynamics and frequency response.
- As a result, the STL axes exhibit alternating (+/– and –/+) behaviour — a dynamic interaction between resistance and flexibility that governs the instrument’s overall response.

This complexity demonstrates that bout-shape behaviour — a primary contributor to acoustic performance — can be enhanced through the use of specific winter-grown beams or bundles with favourable mechanical properties.

Creep behaviour must also be considered. With well-defined STL structures, the locations of creep-related movements can be predicted.

Without STL formations, as in many luthier-made instruments, stress distribution between the bout and C-bout regions becomes unpredictable, leading to unstable acoustic outcomes and uncontrolled long-term deformation.

The spring-like behaviour of the structure must therefore be embraced as the instrument’s internal motor.

The finest instruments display extensive regions with active spring function, indicating that the structure participates directly in sound production.

When structural conditions approach optimality, even minor changes to the varnish — such as rubbing or polishing — can activate new stress behaviours and further influence the acoustic response.

A later section presents the results of a finite-element analysis (FEA) of these plate structures within a complete instrument.

4.7 Dipole Radiation and Constructive Interference

Sound Radiation from Two Structural Sources

The violin does not radiate sound from a single point. Instead, the upper and lower bout structures act as **two spatially separated acoustic sources**, each vibrating with its own characteristic motion. These motions behave as **dipole radiators**, producing alternating pressure zones that “breathe” through the f-holes.

Dipole Behaviour and the F-Hole Interface

As the plates flex, the enclosed air volume is periodically compressed and released. This creates **out-of-phase pressure pulses** at the two f-holes. When the motions of the upper and lower bouts share the **same frequency** and reach the f-holes with **similar amplitude**, their pressure waves combine.

Constructive Interference at Equal Frequency and Level

When two waves of equal frequency and comparable decibel level meet at the f-hole openings, they can reinforce each other. This **constructive interference** increases the effective sound pressure level at specific frequencies, enriching the radiated spectrum.

This dual-source interaction contributes to:

- **Greater projection**, because reinforced frequencies carry more efficiently
- **Enhanced clarity**, as stable interference patterns sharpen partials
- **Increased tonal complexity**, due to the interplay between the two radiating regions

Visual Representation

Figure 4.8 illustrates this principle by showing how the upper and lower bout motions act as paired radiators whose outputs converge at the f-holes, forming zones of constructive reinforcement.

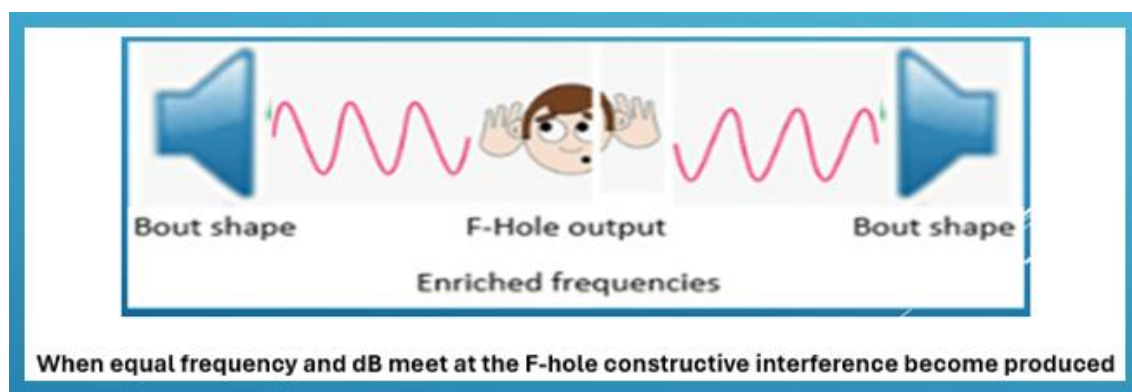


Figure 4.8

This constructive interference depends not only on equal frequency and decibel levels, but also on the **symmetry and timing** established by the STL framework and the tandem behaviour of the top and back plates. When the STL formations are well-defined and balanced across both bout structures, they help ensure that the emitted waves remain **phase-aligned** as they travel outward.

This phase alignment is critical. Even slight asymmetries in STL geometry, beam stiffness, or inflection-line curvature can disturb the timing between the two radiating regions. Such disturbances weaken the interference pattern, reducing projection, tonal clarity, and the instrument's ability to focus energy efficiently.

Thus, the STL framework functions not only as a **mechanical stabilizer** but also as an **acoustic synchronizer**, coordinating the structural and resonant behaviour of the instrument.

Key Points of the Interference Process

- The bout structures on either side emit sinusoidal waveforms.
- These waves converge at the f-hole output, where the listener perceives a blend of enriched frequencies.
- When equal frequency and decibel levels meet at the f-hole, constructive interference is produced.

Between these points, the accompanying figure and explanatory text clarify how the violin's dipole behaviour—driven by STL geometry and beam alignment—creates a focused acoustic output. This convergence is not random; it is the result of a **structurally tuned system** that channels vibrational energy with remarkable efficiency through the instrument's body.

4.8 Motor Function and Its Role in Dipole Enhancement

The quality of the motor system is governed by the **tandem-like interaction** between the buckling of the top plate and the bending of the back.

These two motions are not independent; they operate as a coupled mechanism that drives the instrument's internal pressure cycles.

A central function of this motor is to **increase the dipole behaviour** of the violin, thereby strengthening the “breathing” of the enclosed air volume.

the top plate buckles inward as the back plate bends outward—and vice versa—the alternating pressure fields become more pronounced. This enhances the amplitude of the dipole radiation emerging through the f-holes.

In a well-balanced instrument:

- The top plate's buckling motion provides rapid, high-sensitivity pressure modulation.
- The back plate's bending motion supplies depth, stability, and timing to the pressure cycle.
- Together, they form a synchronized motor that amplifies the alternating pressure pulses responsible for dipole output.

This coordinated behaviour ensures that the air volume does not merely respond passively but participates actively in the radiating process.

The stronger and more coherent the motor action, the more efficiently the violin converts structural motion into **focused acoustic energy**.

Continuation — Plate Stiffness, Varnish Influence, and Structural Sensitivity

Plate stiffness depends on the thickness graduation, which the luthier varies as the arching shape changes.

Varnish, however, is generally applied with an almost uniform thickness. This means that its influence on stiffness is not taken into account in the graduation itself.

As a result, the proportional relationship established by the plate thickness graduation is altered, and the varnish inevitably affects the mechanical outcome of the Motor System's behaviour.

Understanding this makes it possible making adjustment on varnish thickness that will influence the mechanical quality and by that the acoustic result.

The stiffness of each plate directly influences the behaviour and dynamic magnitude of the other, and the motor can function correctly only when both responses are in balance.

For this reason, any improvement must begin with the **top plate**, since it is the top plate that produces the sound and sets the mechanical rhythm for the entire system.

The instrument must be **fully varnished and cured**, because the varnish becomes an integral part of the structural response.

The varnish layer governs the stress behaviour on the outside of the buckling arching structure.

It functions almost like an armour, actively participating in the deformation process rather than merely coating the surface.

As the bout begins to buckle, **tension first develops in the varnish**, long before any significant tension arises in the wood beneath it.

This entire sequence occurs in the bout sectors that buckle outward, indicated by the “+” sign in Figure 4.9.

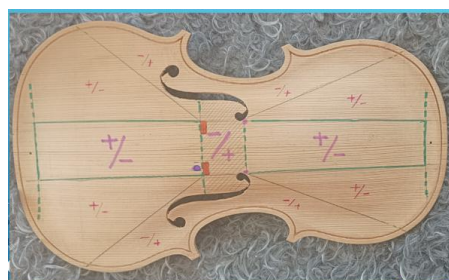


Figure 4.9

Under playing conditions, a **dynamic +/- behaviour** begins to appear.

Specific areas of the bout repeatedly change direction as the structure alternates between outward buckling (“+”) and inward bending (“-”) states, while in the C-bout region the pattern reverses.

These alternating states define the dynamic behaviour of the Motor System and determine how the structure responds under load.

This directional switching is not random; it is driven by the interaction between the static stress conditions of the arching and the dynamic stress conditions produced by the bowing action.

The varnish, acting as a stressed membrane, participates actively in this process and helps define where and how these directional changes occur.

All of this behaviour is directly linked to the **internal framework defined by the STL beams**, which determine how forces are guided, resisted, and redistributed during motion.

On the inside of the arching, the wood simultaneously enters **compression**, creating a counter-zone to the tensioned varnish layer.

The inside of the plate is an **unprotected wood surface**, and because it is not sealed by varnish, it is far more sensitive to changes in humidity.

As moisture levels rise or fall, the inner surface can become softer or harder, altering its stiffness and therefore its contribution to the buckling cycle.

This sensitivity has practical consequences.

A sudden shift from a cold, dry climate—such as Canada—to a hot, humid environment like Bangkok can cause the interior wood to **swell, soften, or shift its mechanical balance**.

Since the motor system depends on precise timing between the buckling of the top plate and the bending of the back, such environmental changes can disturb the instrument’s structural equilibrium.

In other words, the interior surface is not merely passive material; it is a **dynamic participant** in the instrument’s behaviour.

When humidity alters its stiffness, the entire stress distribution changes, influencing:

- the buckling threshold of the arching
- the tension–compression balance around the neutral line

- the responsiveness of the STL framework
- the efficiency of the dipole “breathing” cycle

This is one reason why moving an instrument abruptly between extreme climates can lead to tonal instability or even structural deformation.

The violin is a finely tuned mechanical system, and its internal surfaces—unprotected and moisture-responsive—play a crucial role in maintaining that balance.

Between the tensioned varnish layer and the compressed inner surface lies a **neutral centre line**, a structurally inactive layer that does not participate in the buckling process.

Its presence defines the boundary between tension and compression and helps explain why even small changes in thickness or stiffness can shift the entire stress pattern.

This same structural pattern appears in the **bass bar**: one side of the bar buckles outward while the opposite side is compressed, with a neutral line forming between them.

As described earlier in Chapter 3 (Figure 3.22), even a very small removal of material in a region under tension will alter the buckling response and redistribute stresses throughout the structure.

Because the motor system depends on precise structural motion around the inflection points in front of the end blocks and the STL frame work, even minimal adjustments in the bout and C-bout areas can influence the overall function.

These adjustments affect the dynamic shortening and widening of the instrument during motion, and therefore the efficiency of the dipole cycle.

A further consideration is the **split behaviour of spruce**, which is essential for understanding the internal construction of the wood.

Spruce can split in two principal directions:

- parallel to the winter growth lines
- along the increasing curvature of the arch, forming a cross-over split

The structures between the winter growth lines — the summer wood — align naturally with the **vertical direction of the arching**, contributing to the stability and predictability of the buckling motion.

But in the **horizontal direction**, where the plate aligns with the rib structure, the material layers behave differently, introducing another dimension of structural sensitivity that must be considered in both construction and adjustment.

This is especially true in the **downhill area** from the top toward the end blocks, where the arching level passes through the zone between **8 mm and 3 mm above the rib plane**, while the plate thickness remains approximately **3 mm**.

In this region, the effective length of the wood fibres that connect the layers becomes shorter, reducing the quality of the structural connection and increasing sensitivity to **horizontal shear behaviour**. Figure 4.10

illustrates this relationship.

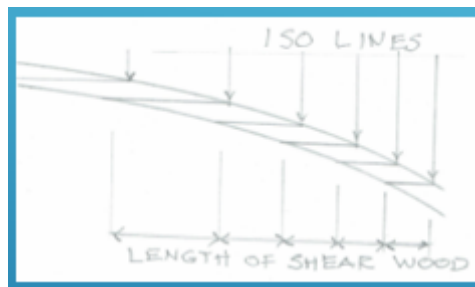


Figure 4.10

As the shear-carrying fibre length shortens, the **buckling-resisting capacity** is reduced and depends primarily on the winter-growth wood that remains intact along the downhill curve.

Note for Understanding — Directional Behaviour During Hollowing

*Luthiers hollow the plate primarily in the **length direction**, and in this orientation the wood behaves very differently from the cross-over direction. Along the grain, the fibres form long, continuous structural paths, and splitting is **far more effective and easier** than across the grain. This means that any reduction of material in the longitudinal direction interacts directly with the natural splitting tendency of spruce.*

Because spruce has limited resistance to shear across the grain, this zone is particularly vulnerable to **horizontal splitting**, a behaviour that directly influences the outcome of the buckling stress forces.

Even a small reduction in stiffness — such as rubbing or thinning the varnish — can alter the local stress distribution and influence both the **buckling transition** and the **stress transformation** in the bout shape that produces the spring function.

The location where this influence is greatest is the area **in front of the end blocks**, where the inflection condition is also found.

As the lower-level structural connections weaken, the tendency for splitting increases, and any adjustment in this region affects both the buckling behaviour and the spring function.

These changes, in turn, influence the resulting **sound quality**.

Thus, the internal structural quality of the wood in this region can be used deliberately to adjust and improve both the stress distribution and the buckling behaviour of the beam function, and thereby the function of the bout area.

When the conditions change — for example, by rubbing the bout structure — the resulting increase in **bulging** influences the cross-over bout shape and contributes to the formation of the **inflection** that follows the outline.

The winter-growth wood at the lower level forms a **shorter arc** from the outline toward the STL structure.

It is important that these structures also undergo buckling so that the essential **spring quality** is maintained.

There can be no doubt that the configuration of the **ISO line layout** has a significant impact on the resulting behaviour of the structure.

As the effective fibre-to-fibre connection becomes progressively shorter downhill, the quality of the structural connection decreases. *Figure 4.9* illustrates how the shear structural length shortens in the downhill direction.

The red lines indicate the regions that carry the shear stress.

As the fibre-to-fibre connection weakens, the material becomes increasingly sensitive to changes in tension, varnish thickness, and the buckling stresses that arise.

These structural wood qualities are most pronounced just above the higher level of the **inflection point** along the centre line beneath the fingerboard.

If we reduce thickness in such a region — or even reduce only the thickness of the varnish — the buckling stress condition changes, and with it the dynamic behaviour of the motor system.

This means that the internal structural quality of the wood and the varnish can be used deliberately to adjust and improve both the **buckling behaviour** and the **spring function** of the plate, and ultimately the **quality of sound production**.

4.9 Adjustment Procedure starts at the upper bout under the finger board

To improve the motor system's activation, the following procedure must be carried out carefully and step by step:

1. Start with a fully varnished instrument, where the varnish is cured and capable of being rubbed.
 - If the instrument has an oil varnish, rubbing can be performed using any suitable polishing medium.
 - The author's preferred medium is copper polish.
2. Prepare a soft woven cloth, moistened slightly.
3. Insert a thin stick under the fingerboard to support access.
4. Gently rub at the presumed inflection point in front of the end blocks especially focusing on the bout side.
5. Allow the instrument to rest for approximately two hours, so the structure can deflect and redistribute stress within the buckling beam system.
6. Test the instrument's response, ideally by recording with Audacity:
 - Play the open G-string with no additional pressure.
 - Use a bow stroke of approximately five seconds.
7. Evaluate the frequency response:
 - If improvement is observed, proceed to the next adjustment step.
 - If no improvement is observed, repeat the rubbing process, potentially extending it further along the bout.
8. If improvement still does not occur, the issue may lie in the **tandem function** between the top and back plates:
 - In this case, rubbing must be performed on the **buckling beams** and, if necessary, on the **centre of the back plate**, at the highest point of the length-curve where bending forces are most active.
 - A **wider area** of the bout structure may be treated, since the back plate is not restricted by the width limitations imposed by the upper f-holes.

It is essential to understand that **buckling in the top-plate bout shape works in direct tandem with bending in the back plate**.

These three structures—the **top plate**, the **buckling beams**, and the **central bending region of the back plate**—must be brought into balance.

This requires the luthier first to *hear* the effect of rubbing the varnish, but once that initial reference is established, the process can be evaluated more objectively by recording the open G-string.

The resulting graph displays the acoustic outcome directly, allowing the luthier to monitor how each adjustment influences the behaviour of the Motor System.

When their behaviour is aligned, the **motor function** strengthens and the **dynamic response** of the instrument becomes more efficient.

Once the rubbing procedure produces a clear improvement, the motor system has likely reached a condition in which it can **activate other frequencies more easily**. This indicates that the structural balance between buckling and bending has been restored to a level where the instrument can respond more freely across its frequency range.

This improvement should be audible and measurable, and it prepares the instrument for further functional refinement.

4.10 Understanding the Structural Response After Adjustment

Once a rubbing procedure has been carried out and the instrument has been allowed to rest, the structure enters a brief period of **stress redistribution**.

During this phase, the buckling-beam system adjusts itself to the new stress conditions created in the varnish, and these altered conditions establish a **new and permanent buckling response** in the plate.

As a result, the **stress distribution and the spring function change permanently** as the system settles into its new equilibrium.

The first indication of successful adjustment is that a **more responsive motor system**, in which the tandem action between top-plate buckling and back-plate bending becomes easier to activate. This is typically observed as:

- quicker onset of vibration,
- clearer articulation at low bow pressure,
- and improved stability of the open G string, which serves as the primary indicator of motor activation.

Because the deformation involved is extremely small — well below **0.05 mm** — the improvement does not result from large geometric changes, but from **micro-adjustments in stress pathways**.

If the rubbing has been effective, the instrument will exhibit:

- a more even distribution of buckling stress,
- reduced resistance at the bout inflection zones,
- and a more coherent transfer of energy from bridge to body.

If the improvement is only partial, or if the response remains unstable, this indicates that the tandem function between top and back is still unbalanced.

In such cases, further refinement may be required — either by extending the treated area or by adjusting the back plate.

The purpose of this step is not to force deformation, but to **activate the natural mobility of the motor system**, allowing the instrument to operate within its optimal structural window.

Rubbing the back plate in the corresponding location to the top plate also influences the **buckling conditions of the top plate**.

When the back plate is allowed to bend more freely, the structural balance between the two plates shifts.

Increased bending capacity in the back means that the top plate must carry a greater share of the buckling load, which raises the **stress condition** in the top plate.

This altered load distribution establishes a **new and permanent spring function**, because the tandem relationship between buckling in the top and bending in the back has been recalibrated.

The spring quality immediately tells the luthier the acoustic output level produced by a specific bowing action.

In practical terms, the motor system becomes more responsive when this balance is improved, and the structural change itself is not temporary — it represents a **new equilibrium** in the instrument's mechanical behaviour.

Therefore, when rubbing the top plate does not produce the expected result, the luthier must also treat the **corresponding region of the back plate**.

The two plates operate as a coupled system, and the motor function cannot be fully optimised if only one side of the tandem mechanism is adjusted.

It is important to keep in mind that there is no natural vibrational “driving track” running from the bridge feet to a specific structural element capable of initiating a frequency response at a predetermined location.

Such a mechanism would be illogical: a region of the plate would have to change its vibrational track 15–20 times per second in different directions to follow the bridge-foot frequency, tracing a precise path across the arching to generate sound.

The vibrational distance from the upper f-hole to the adjacent structures is far too complex for such a direct dynamic track to exist.

Instead, the **frequency response arises spontaneously** from structures that are already in a dynamic state produced by the basic motor activity.

Structures on the arching begin vibrating and generate specific frequencies with high amplitude only when the **local stress condition in the bout shapes aligns with the played frequency**.

In other words, the instrument does not “follow” the frequency — the frequency emerges when the structural conditions are right.

It is important to understand that we are dealing with a **mechanical process** based on **static stress conditions** that, when activated by the bowing action on the strings, produces **dynamic structural stress conditions** which respond directly to the frequency input delivered through the bridge feet.

The STL framework determines the **quality and efficiency** of this mechanical process by defining how forces are guided, transformed, and redistributed within the plates

In other words, the STL structure does not create the vibration — it predicts how well the instrument is able to **convert mechanical stress into dynamic response** when excited by the played frequency.

4.11 Result of the rubbing.

A continuing process within the plate system spontaneous activates the adjacent beam-like structures that contribute to higher-frequency responses.

In these regions, the effectiveness of the response depends on the sensitivity of their spring function.

When this spring behaviour is limited, the adjacent structures do not fully participate in the higher-frequency activation.

The curve length and chord length of each winter-grown beam diminish gradually along its span.

This gradual reduction increases the local stiffness in the buckling zone between the STL and the inflection points for each individual beam.

As stiffness increases, the spring sensitivity of these areas becomes more restricted.

Practical observations show that rubbing these adjacent bout beams reduces their stiffness, allowing the spring function to become more sensitive and more responsive to bowing action. During playing, the motor system must therefore also engage the spring function of these neighbouring bout structures so that they contribute effectively to the instrument’s dynamic behaviour.

In this way, the instrument's response is governed not by a forced vibrational pathway, but by the internal readiness of each structural zone to participate when its stress state matches the excitation.

4.12 The visualized frequency result

The result observed when playing the open G string now verify the result of the rubbing and is therefore a verification of structural quality: the same structures will activate with high dB when their specific frequencies are played on other strings.

The activation is governed by the **stress-dependent readiness** of the structure to respond not only from one bout shape.

Figure 4.7.1 and 4.7.2 shows this clearly on a well-trimmed violin. We observe structures producing **higher frequencies than the played tone** (the open G string), demonstrating that the system is not simply following the input frequency.

What we see is the result of **dipole and interference function** in a well-balanced arching shape — a structure capable of producing multiple frequencies with high precision, all with nearly equal dB levels.

When two regions of the arching produce the same frequency through the dipole spring function, interference arises.

This interference is not caused by wave collision, but by **structural synchronisation**: two bout regions responding in phase, reinforcing each other and amplifying the vibrational energy at that frequency.

The spectral result is a peak with unusually high stability and precision, confirming that both bout regions are dynamically aligned.

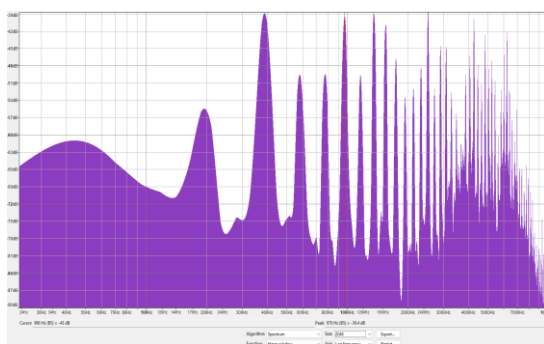


Figure 4.7.1

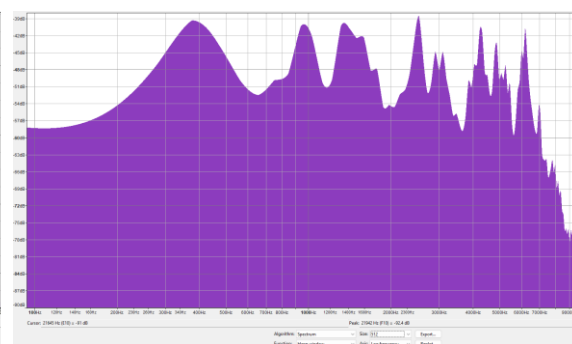


Figure 4.7.2

This is remarkable, and it confirms that the quality of the arching shape — in both the upper and lower bout regions — work in **dipole behaviour**, determines the instrument's ability to generate a coherent and finely balanced spectrum of frequencies.

What we see in the spectral graph is not a classic “interference pattern” in the wave-physics sense (like overlapping sine waves), but rather the **result of a dipole spring function** interacting with the motor system. The graph shows:

- **Multiple frequency peaks** with nearly equal dB levels
- **Higher-frequency activation** even when only the open G string is played
- A **balanced spectral distribution**, suggesting that the arching shape enables spontaneous activation across a wide frequency range

4.14 More About Moisture and function

Moisture influences the behaviour of the wood structure and therefore affects the stress conditions that govern buckling behaviour, particularly the quality of the spring function.

As the moisture content changes, the stiffness of the winter-grown beams and the adjacent structures shifts accordingly.

Different regions of the plate respond with varying sensitivity, altering local stress conditions and modifying the dynamic behaviour of the system.

Over time, moisture-related variations inevitably influence both the type and the magnitude of creep that develops within the plate structure.

These long-term changes gradually redistribute internal stresses and slowly reshape the arching of the instrument but not the STL framework

This process is not inherently negative.

The gradual evolution of stress conditions allows the structure to settle into a state that supports the wide range of dynamic behaviours characteristic of a great instrument.

The old master instruments — now 300 to 400 years old — have all undergone this long transformation, and we have no direct knowledge of their original structural shape or stress condition.

The STL framework plays a decisive role in how the structure responds under moisture- and creep-driven conditions, guiding the redistribution of stresses and influencing how the arching evolves over time.

For this reason, luthiers attempting to copy an old master instrument must keep these factors in mind.

Replicating an arching shape alone cannot reproduce the behaviour of the great masters' instruments without understanding the underlying structural principles — some of which are revealed in this work.

4.15 Comparative Frequency Results of STL-Based Violins

The three violins presented at the beginning of this chapter all exhibit perfect STL conditions in both their exterior and interior structures.

The frequency results demonstrate that all three instruments produce almost identical acoustic responses.

In each case, the characteristic **Bridge Hill (BH)** peak and the associated interference frequencies are clearly visible.

By comparing these spectra, it becomes clear that the **STL-controlled structural framework** is the determining factor behind the instruments' nearly identical frequency behaviour.

The consistency of the Bridge Hill peak, the interference frequencies, and the overall spectral balance confirms that when the STL geometry is correct, the violin's motor system and dipole structures operate in a predictable and highly coordinated way.

Such uniformity is possible only when the arching shapes are governed by the **structural framework defined by the STLs**.

This framework ensures that the motor system, beam alignment, and bout structures respond in a stable and repeatable manner across different instruments.

According to Prof. E. Jansson (KTH, Stockholm), playing the open G string and recording the result in Audacity provides the clearest indication of an instrument's capacity across its entire frequency range.

This simple procedure reveals the full frequency spectrum, as demonstrated in the images presented in this paper.

4.16 Neck Heel Problems That May Influence the Motor System Result

The structural quality of the neck — particularly the curvature of the neck heel — can significantly influence the dynamic behaviour of the instrument.

Although the heel shape is largely standardised across instruments, its curvature still generates a bending moment that must be carried by the fibres in this region.

When the neck heel begins to show signs of fibre separation, a secondary dynamic behaviour can develop within the fibre structure itself.

Once this behaviour is activated, an adverse interaction may arise between the neck and the buckling bout shape. This interaction reduces the amplitude of the dynamic buckling behaviour in the bouts.

The technical consequence is a measurable reduction in the dynamic spring function of the system. As the spring function weakens, the instrument may lose clarity, responsiveness, and overall dynamic consistency.

These conditions are not directly measurable, yet their influence becomes evident through a reduction in sound quality.

In cello construction, luthiers have attempted to mitigate similar neck-heel problems by inserting a dowel inside the neck to prevent fibre separation and stabilise the structural behaviour.

The red line marks the regions where fibre-split structures may occur.

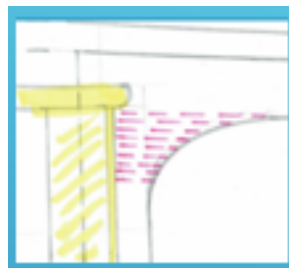


Figure 4.10

As a test, the stiffness at the neck heel was reduced by scraping the curve of the heel. This strongly affected the dynamic behaviour of the body, and the tone quality deteriorated noticeably. Energy was lost in the neck heel, which became dynamically active instead of maintaining its role of holding and distributing stress.

The true solution was to re-graduate the stiffness of the entire structure by reducing the varnish thickness across the whole instrument. Only then did the system return to balanced behaviour.

4.17 The Bridge Functions in the Motor System

The bridge plays a crucial dual role in the violin's operation:

1. **Motor System Component** – It initiates and drives the instrument's complex mechanical and dynamical response within the three-beam system.
2. **Frequency Provider** – It delivers the input frequencies from the strings and, through spontaneous frequency reaction, activates the structures in the bout regions.

3. **Pre-Activated Structures** – These bout structures are already in a dynamic state when the beam system is set into motion, allowing the spring function to develop immediately and efficiently.
4. **Dual Behaviour System** – The violin therefore operates as a dual-behaviour system: the central beam structures act as the driving motor, while the string–bridge assembly functions as the frequency supporter.

This duality ensures that the bridge not only energises the motor system but also shapes and defines the tonal spectrum available to the player.

4.18 STL Symmetry and Arching Coordination

The effectiveness of dipole radiation and constructive interference depends on more than beam alignment alone — it requires structural symmetry across the STL framework.

- When the arching geometry is coordinated with the STL formations, the left and right bout structures can emit waves that remain phase-aligned across a wide frequency range.
- This symmetry ensures that constructive interference is not restricted to isolated frequencies but becomes a system-wide behaviour, enhancing tonal richness and projection.
- In contrast, asymmetrical STL geometry or uneven arching can lead to phase cancellation, reducing clarity and weakening the instrument's dynamic response.

STL symmetry is therefore not merely a geometric preference — it is a functional necessity for achieving consistent acoustic performance.

The STL framework plays a central role in maintaining phase alignment between the left and right bout structures. When the geometry is symmetrical and the beam bundles are coordinated, the emitted waves can remain in phase across a broad frequency range, supporting sustained constructive interference at the F-hole output. Even minor asymmetries — whether in arching, beam stiffness, or STL layout — can disturb this balance, leading to phase cancellation and diminished tonal projection.

In this way, STL symmetry stands not only as a structural ideal but as a fundamental requirement for reliable acoustic behaviour.

4.19 Frequency Spectrum Highlights

- **Fundamental & Harmonics**
 - The open G-string fundamental lies at ~196 Hz.
 - Harmonics appear at ~400, 600, 800, and 1000 Hz, forming a strong harmonic series that indicates efficient coupling between string and body.
- **Bridge Hill Region (~2000 Hz)**
 - This is a well-known resonance zone in violins where the bridge and body amplify higher frequencies.
 - A strong response here suggests that the instrument's geometry and STL-guided dynamics are functioning optimally.
- **Constructional Interference Peak (~4000 Hz)**
 - This higher-frequency feature reflects complex interactions between body modes and string excitation.
 - It can add brilliance to the sound but requires balance to avoid harshness.

Together, these spectral highlights demonstrate how **STL formations and beam coordination** shape the violin's acoustic output — from fundamental coupling to high-frequency brilliance — ensuring both stability and tonal richness.

4.20 Structural Interpretation

- The **coherent spectral shape** implies that the instrument's geometry, material stiffness, and STL (structural logic) are properly aligned.
- When the system is near optimal, even small surface interactions — such as **varnish rubbing** or **micro-adjustments** — can noticeably affect frequency resonance and sound projection.
- This sensitivity arises because the instrument is already constructed to respond dynamically; it requires only **fine-tuning** rather than major structural change.
- As a result, even subtle modifications can shift the balance of **energy transmission**, influencing the acoustic outcome in perceptible ways.

In essence, the violin's spectral coherence reveals a system poised for **dynamic participation in sound production**, where minimal interventions can unlock significant changes in tonal character and projection.

Figure 4.8.1 and 4.8.2 show the spectrum recorded and visualized of the played open G-string, making the structural-acoustic interactions directly observable.

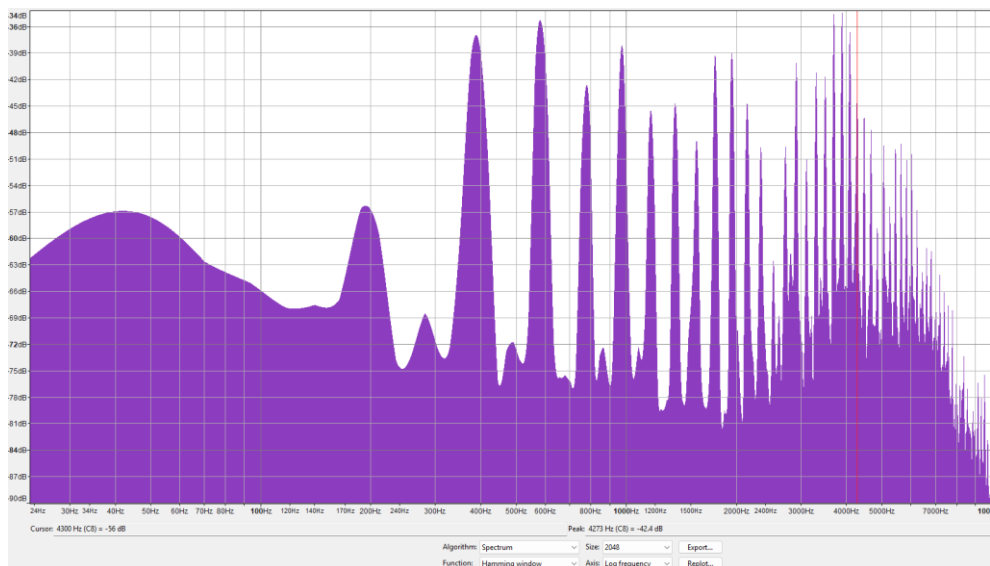


Figure 4.8.1

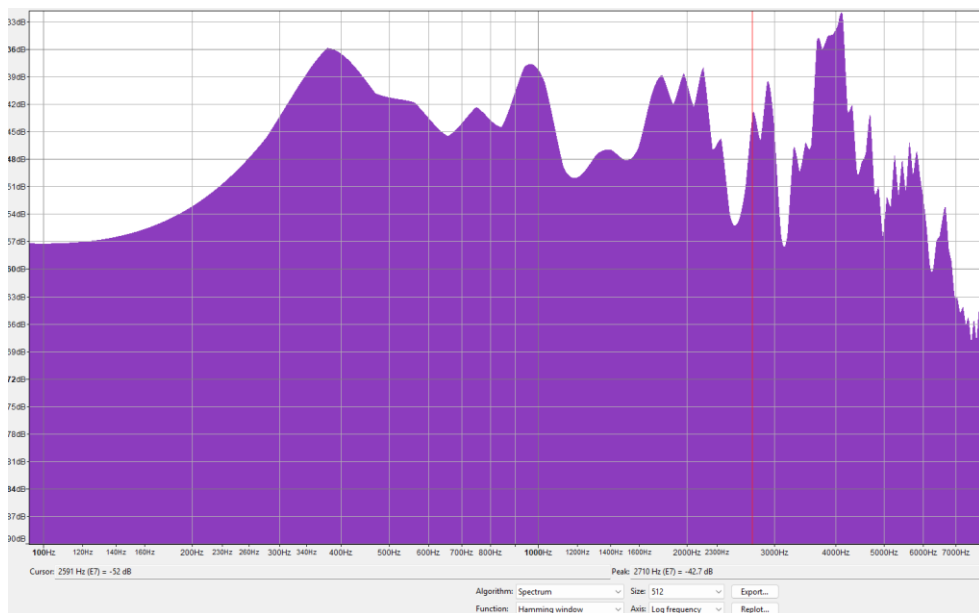


Figure 4.8.2

4.21 Creep Behaviour and Long-Term Stability: Findings from a Master's Dissertation on Real Plates

In cooperation with the Division of Structural Mechanics at LTH, Lund University, Sweden, a master's dissertation based on real instrument plates — conducted with input from the author — demonstrates that the STL framework retains its structural condition under

sustained load. The study was carried out by Erik Tunlid and Joel Väreälä under the supervision of Professor Kent Persson.

Using violin drawings and structural data provided by the author, a finite-element model of a complete violin was created and tuned by tensioning the strings.

The model incorporates non-linear geometry to account for the effects of string tension, and a time-hardening creep model was implemented to study the long-term ageing process of violins.

The complete dissertation is available at:

<https://www.byggmek.lth.se/fileadmin/byggnadsmekanik/publications/tvsm5000/web5247.pdf>

While the frequency spectrum reveals the violin's immediate acoustic response, its long-term stability depends on how the structure manages creep behaviour — the gradual deformation of wood and glue joints under sustained stress.

Understanding this behaviour is essential for evaluating how the STL framework preserves its functional geometry over time and how the instrument maintains its dynamic capacity across decades and centuries.

Figures 4.9.1 and 4.9.2 show the two plates used in the structural model for the FE analysis. Both plates exhibit perfect STL conditions on both the inside and outside surfaces.



Figure 4.9.1 inside STL



Figure 4.9.2 outside STL

Figure 4.10.1 and 4.10.2 show the thicknesses of the plates

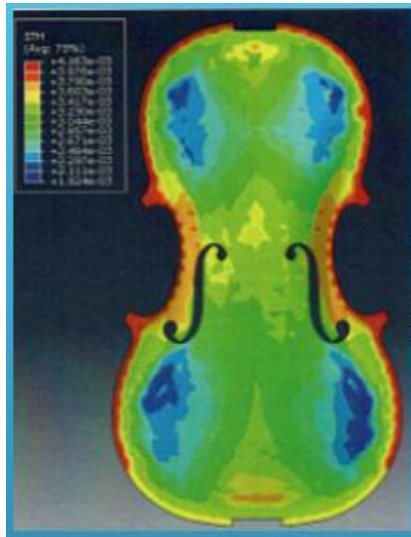


Figure 4.10.1

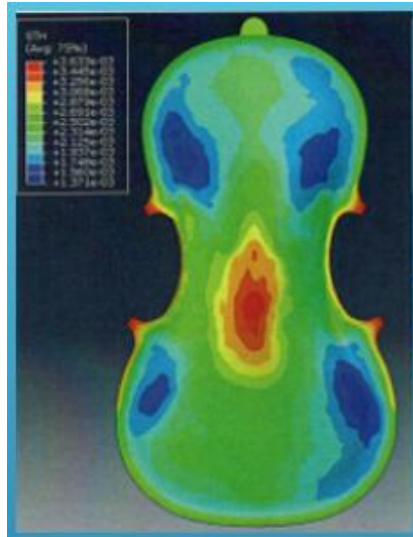


Figure 4.10.2

Over time, the STL framework and its beam bundles must resist slow geometric shifts that could otherwise disturb phase alignment, resonance balance, and tonal projection.

This endurance arises from the interplay of material stiffness, arching geometry, and structural logic, which together safeguard the instrument's acoustic integrity.

Because the violin is designed to respond dynamically, even small changes in load distribution can accumulate over time.

If left unchecked, these shifts may weaken constructive interference or alter the balance of energy transmission.

Understanding creep behaviour is therefore essential: it reveals how the violin maintains its functional equilibrium across decades of performance, ensuring that the acoustic qualities demonstrated in the spectrum remain stable throughout the instrument's lifespan.

As explained earlier, the STL framework does not deflect. This evidence shows that creep does not affect the STL geometry, confirming its role as a stable backbone within the instrument.

While other regions may undergo gradual deformation, the STL framework remains intact, ensuring long-term stability and preserving the acoustic logic of the violin.

4.22 Creep Results found by FEA

Findings show that creep affects the STL formations only minimally, confirming their role as a stable backbone within the instrument. While other regions of the plates may experience gradual deformation, the STL framework remains intact, ensuring long-term stability and preserving the acoustic logic of the violin.

4.23 Creep Strain Over Time

Figure 4.11 illustrates the development of creep strain in the violin's top plate after 30 days, 2 years, and 100 years.

The bold red lines mark the STL framework on the surface, including its extension to the rib, and show that this structure remains intact despite gradual deformation in the surrounding regions.

This visualisation confirms that, although creep affects the plate material over long periods, the STL formations are only minimally influenced.

Their stability ensures that the instrument's structural logic and acoustic transmission remain preserved across decades of performance.

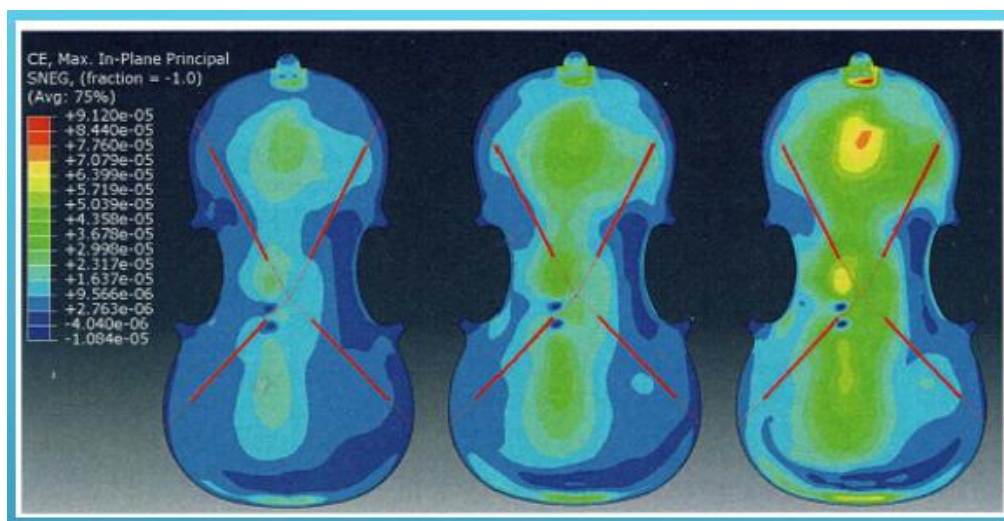


Figure 4.11

Creep Strain in the Back Plate Over Time

Figure 4.11 illustrates the development of creep strain in the violin's back plate after 30 days, 1 year, and 100 years.

The bold red lines mark the STL framework, including its extension to the rib, and show that this structure remains intact despite gradual deformation in the surrounding regions.

This visualisation confirms that, although creep affects the plate material over long durations, the STL formations are only minimally influenced.

Their stability preserves their role as a structural backbone for acoustic transmission and ensures long-term integrity within the instrument.

Creep Strain in the Top Plate Over Time

Figure 4.12 illustrates the development of creep strain in the violin's top plate after 30 days, 2 years, and 100 years.

The **red lines mark the STL framework**, which remains structurally intact despite gradual deformation in surrounding regions.

This visualization confirms that while creep affects the plate material over long durations, the STL formations are only minimally influenced.

Their stability ensures that the instrument's **structural logic and acoustic transmission** remain preserved across decades of performance.

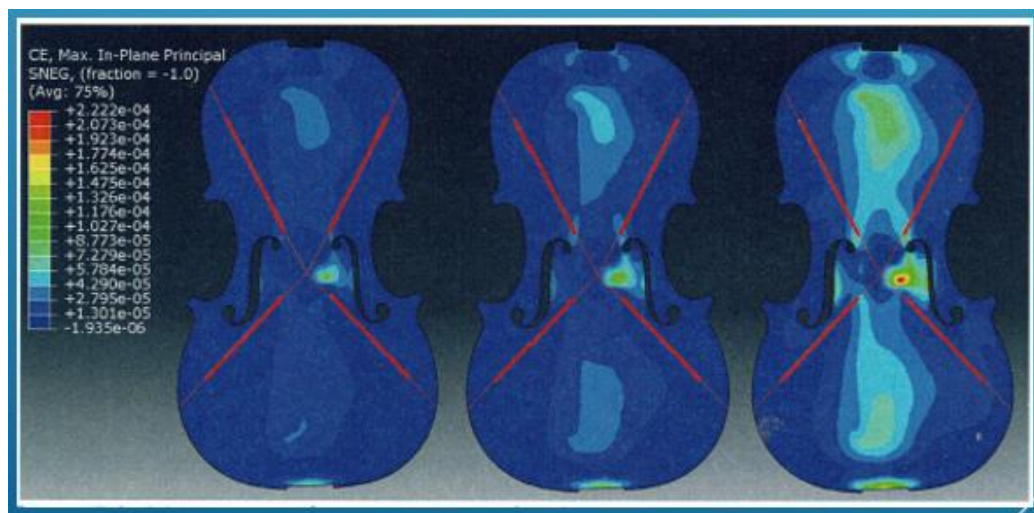


Figure 4.12

Figure 4.12 illustrates the development of creep strain in the violin's top plate after 30 days, 1 year, and 100 years.

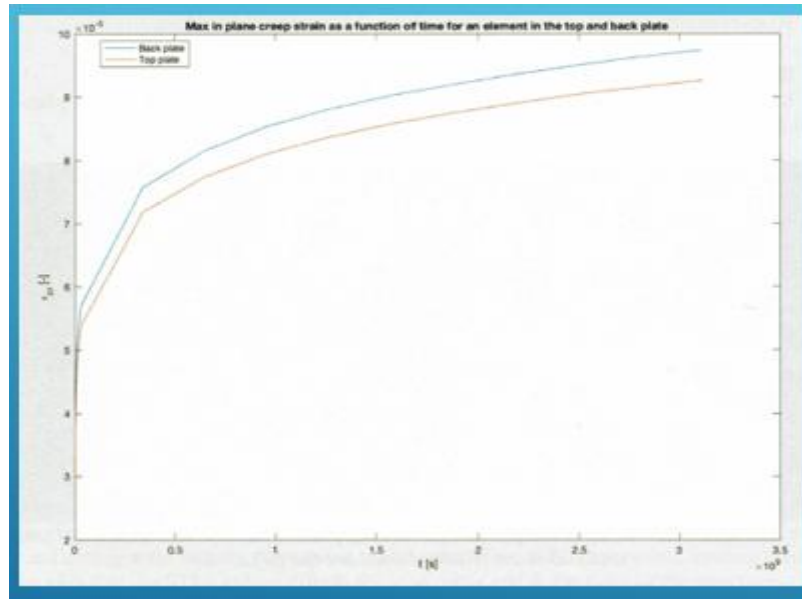


Figure 4.13

The Long term creep deformation is shown on figure 4.13, top red line, back blue line.

With this technical information established, it becomes possible for the luthier to improve the instrument in its **playable condition, even after varnishing**.

Final Word

What has been explained in these four chapters will undoubtedly be controversial for both scientists and luthiers. The prevailing view — shared by many researchers and makers, including those associated with the VSA — is that the violin is driven primarily through the bridge, and that the end blocks remain essentially static.

However, their interpretation is impossible, because the supporting structures of a real instrument cannot behave in the manner they propose; these structures must stabilise the instrument, not deform as their model assumes. Figures 1.2.1 to 1.2.6 illustrate this clearly. In their model, vibrational energy flows downward from the bridge feet into the top plate, while the blocks act as fixed boundaries that do not participate in the dynamic behaviour.

Their understanding is incomplete because it excludes essential structural mechanisms, making it incapable of describing the instrument's actual behaviour.

The bout regions do not respond because energy is mechanically “sent” to them along a vibrational track from the bridge — nor do they respond because of any action from the end blocks, which are not the driving mechanism of the instrument. Instead, they respond because the local stress conditions created by the motor system place them in

a state of structural predisposition. When the played frequency aligns with these stress conditions, the structure becomes activated spontaneously.

It is essential to include both the upper and lower bout regions, which participate through the dipole spring function. This means that the end block regions cannot be considered fully static; they are part of a coordinated structural system whose mobility is necessary for producing coherent and balanced dynamical behaviour.

To understand why this must be so, we must consider the basic idea the original inventor of the violin likely had in mind. The geometry of the arching and the proportional relationship between the upper and lower bouts indicate an intention to create a self-activating structural system. The violin was conceived as a body in which the spring function of the arching could operate freely, allowing the instrument to respond spontaneously when local stress conditions aligned with the played frequency.

The inventor's insight was not to force vibration along a mechanical track, but to create a structure that would activate itself when placed under the correct stress conditions.

Figures 4.8.1 and 4.8.2 illustrate this behaviour clearly. We observe structures producing frequencies higher than the played tone (the open G string), demonstrating that the system is not simply following the input frequency. What we see is the result of the dipole spring function in a well-balanced arching shape — a structure capable of producing multiple frequencies with high precision, all with nearly equal dB levels.

When two regions of the arching produce the same frequency through the dipole spring function, interference arises. This interference is not caused by wave collision, but by structural synchronisation: two bout regions responding in phase, reinforcing one another and amplifying the vibrational energy at that frequency. The spectral result is a peak with unusually high stability and precision, confirming that both bout regions are dynamically aligned.

This is remarkable, and it confirms that the quality of the arching shape — in both the upper and lower bout regions — working in dipole spring behaviour, determines the instrument's ability to generate a coherent and finely balanced spectrum of frequencies.

Ultimately, only a well-balanced arching operating in dipole spring behaviour can explain the violin's ability to produce a coherent and finely structured spectrum of frequencies.

Note:

The three violins presented were made by violist and luthier Paulo Parlagreco, São Paulo, Brazil, working in collaboration with the author and following the author's technical direction.

